

DEMAND DYNAMICS

Photovoltaics account for over 30% of global silver demand, making the sector sensitive to price movements in the white metal. Copper has wider use in solar module manufacturing and wind energy with solar accounting for 51% demand in FY25 compared to 30% in FY23, according to the International Copper Association India. Even though prices of both these metals have dropped this month, the mood of the industry is sombre.

“Silver paste typically represents 10-15% of photovoltaic cell manufacturing costs, and sustained increases can place pressure on margins and near-term project economics across the value chain,” Gagan Channana, Joint MD & CEO of solar panel manufacturer Jakson Solar Modules and Cells, tells *Business Today*.

Solar is set to be the largest contributor to India’s 500GW non-fossil fuel energy target by 2030 with a total capacity of 300GW compared to 135 GW at present. Wind energy is set to double by 2030 from 54GW at present.

Union Budget 2026-27 boosted solar energy schemes with an allocation of ₹30,539.36 crore—a 32% increase over FY26. A major highlight of the Budget is the continued expansion of the *PM Surya Ghar: Muft Bijli Yojana* with an allocation of ₹22,000 crore for rooftop solar.

According to the World Silver Survey 2025, solar photovoltaics remain one of the largest and most structurally important sources of industrial silver demand, accounting for a substantial share of electrical and electronics consumption. Silver is pivotal in solar cell production due to its conductivity.

Experts say the demand for silver is set to rise in India with several industrial houses announcing big investments in solar photovoltaic cell manufacturing to reduce dependence on China. Reliance Industries is setting up a mammoth 10 GW annual solar manufacturing facility in Gujarat. Tata Power has commenced commercial production of 2GW solar cells at its facility in Tirunelveli, Tamil Nadu. Adani, Jackson, Waare Energies, GREW Solar, and Saatvik Greens are also expanding capacity.

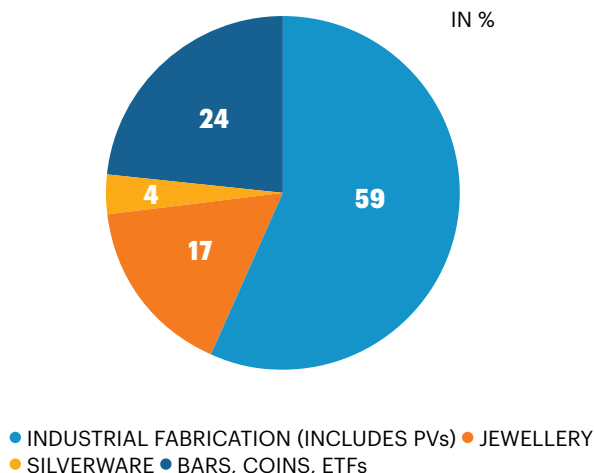
METALLIC CHALLENGE

Not just silver. Rising copper and aluminium prices are also bothering renewable energy sector players. Domestic copper demand grew 9.3% year-on-year in FY25.

With India targeting near doubling of the installed solar capacity, the focus will be on ramping up solar

GLOBAL SILVER DEMAND

► Photovoltaics account for a big chunk of silver demand, under the category of Industrial Fabrication



SOURCE WSS: WORLD SILVER SURVEY 2024

photovoltaic module manufacturing capacity, adding to rise in demand for these metals. India’s photovoltaic manufacturing capacity stands at 100 GW, but the country is still largely dependent on China for solar cells and modules.

Large amounts of copper is used in module manufacturing while solar cells use silver in large quantities. Vinay Thadani, Director & CEO of GREW Solar, another solar photovoltaic module manufacturer, says silver is a critical input in solar photovoltaic cell metallisation, and its sharp price appreciation has added measurable pressure.

“From an industry perspective, the increase in silver prices alone can add ₹4,00,000-5,00,000 per MW to solar photovoltaic manufacturing costs, depending on cell technology and silver intensity. While this tightens margins in the near term, manufacturers are increasingly offsetting a part of this impact through efficiency improvements and material optimisation rather than passing costs directly to customers,” Thadani tells *BT*.

To add wind to the fire, silver licensing by China